ELM361 Yaz 2021 AS Çözümleri

① $H(f) = \frac{2}{\sqrt{1+(0{,}001f)^2}} e^{-j\arctan(0{,}001f)} \Rightarrow |H(f)| = \frac{2}{\sqrt{1+(f/1000)^2}}$

$\theta_h(f) = -\arctan(f/1000)$

(a)

$|H(0)| = 2$

$t_d(f) = -\frac{1}{2\pi}\frac{d\theta_h(f)}{df} = \frac{1}{2\pi}\frac{1000}{1000^2+f^2} \Rightarrow t_d(0) = \frac{1}{2\pi\times 1000} \cong 1{,}6\times10^{-4}$

$|H(f_0)| \geq 0{,}95|H(0)| = 0{,}95\times 2$

$\frac{2}{\sqrt{1+(0{,}001f_0)^2}} \geq 0{,}95\times 2 \Rightarrow f_0 \leq 328\ Hz$

$t_d(f_0) \geq 0{,}95\, t_d(0) = 0{,}95\times\frac{1}{2\pi\times1000} \Rightarrow \frac{1}{2\pi}\frac{1000}{1000^2+f^2} \geq 0{,}95\times\frac{1}{2\pi\times1000}$

$f_0 \leq 229\ Hz \Rightarrow \boxed{B = f_0 = 229\ Hz}$

Bu bandda $y(t) \cong K.x(t-t_d) = 2\,x(t-1{,}6\times10^{-4})$

(b) EYF: $\mathcal{E}_h(f) = |H(f)|^2 = \frac{4}{1+(f/1000)^2} \xrightarrow{\mathcal{F}^{-1}} R_h(\tau)$ özilişki f.

$\mathcal{E}_h(f) = \frac{4}{1+\left(\frac{2\pi f}{2\pi 1000}\right)^2} = \frac{4\times(2\pi\times1000)^2}{(2\pi\times1000)^2+(2\pi f)^2} = \frac{4000\pi \quad 2\times\overbrace{2\pi\times1000}^{a}}{\underbrace{(2\pi\times1000)^2}_{a}+(2\pi f)^2}$

$\xrightarrow{\mathcal{F}^{-1}} R_h(\tau) = 4000\pi\cdot e^{-2000\pi|\tau|}$ $\quad \left(e^{-a|\tau|},\ a>0 \overset{\mathcal{F}}{\leftrightarrow} \frac{2a}{a^2+(2\pi f)^2}\right)$

$E_h = R_h(0) = 4000\pi\ J$

(c) $x(t) = 10\cos(400\pi t) + 20\sin(800\pi t)$

$= \underbrace{5}_{x_1} e^{j2\pi 200t} + \underbrace{5}_{x_{-1}} e^{-j2\pi 200t} + \underbrace{\frac{10}{j}}_{x_2} e^{j2\pi 400t} \underbrace{-\frac{10}{j}}_{x_{-2}} e^{-j2\pi 400t}$

$S_x(f) = \sum_n |x_n|^2 \delta(f-nf_0) = 25\,\delta(f-200) + 25\,\delta(f+200) + 100\,\delta(f-400) + 100\,\delta(f+400)$

$S_y(f) = S_x(f)|H(f)|^2 = \sum_n |x_n|^2 |H(nf_0)|^2 \delta(f-nf_0)$

$P_y = \int_{-\infty}^{+\infty} S_y(f)\,df = \sum_n |x_n|^2 |H(nf_0)|^2$

$= 25\,|H(200)|^2 + 25\,|H(-200)|^2 + 100\,|H(400)|^2 + 100\,|H(-400)|^2$

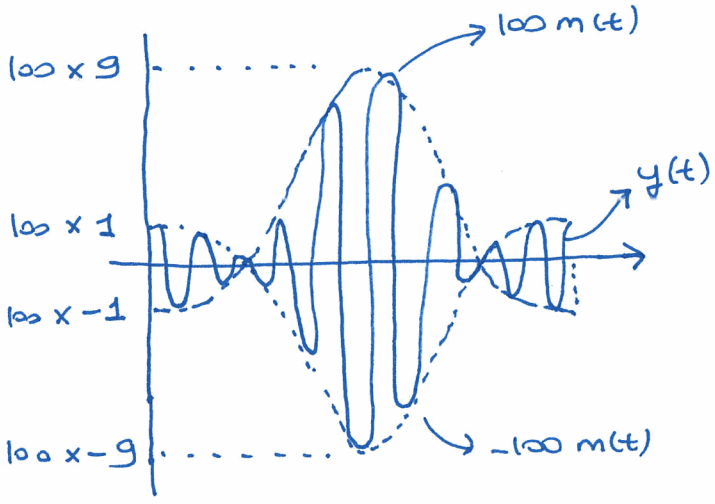
$= 2\times25\times\frac{4}{1+(1/5)^2} + 2\times100\times\frac{4}{1+(2/5)^2} = 881{,}96 \cong 882\ W$

② $m(t) = 4 - 5\cos(1000\pi t)$, $c(t) = 100\cos(10\,000\pi t)$ DSB-SC-AM (TB-GYB-GM)

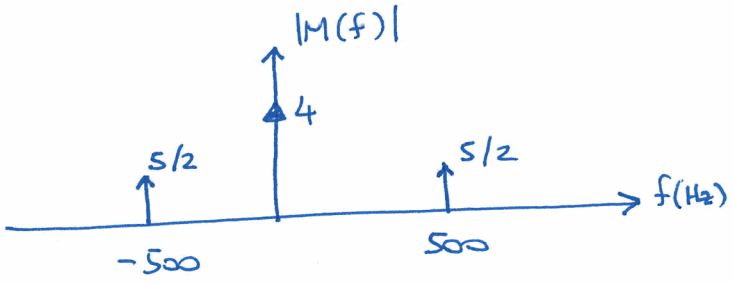
$B = 500$ Hz, A_c, $f_c = 5$ kHz

a) $y(t) = m(t).c(t) = 100\,[4 - 5\cos(1000\pi t)]\cos(10\,000\pi t)$

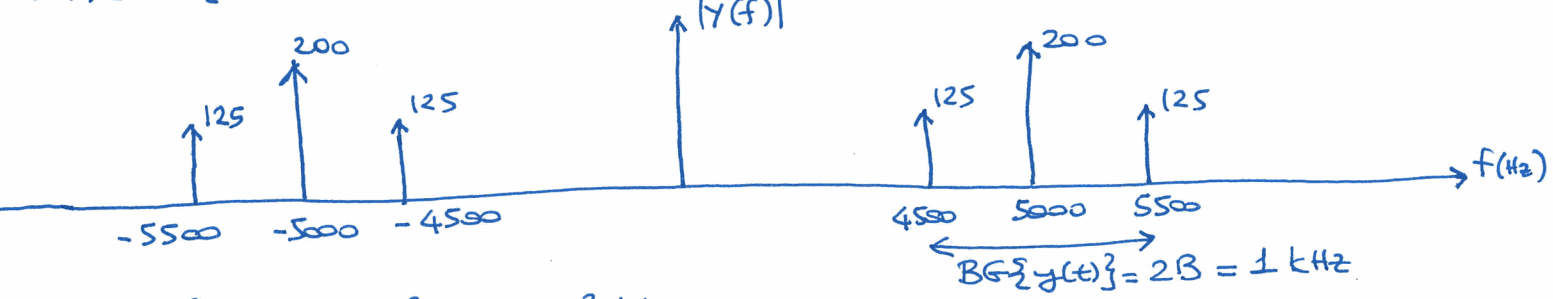
$= 400\cos(10\,000\pi t) - 500\,\frac{1}{2}\,[\cos(9000\pi t) + \cos(11\,000\pi t)]$



$M(f) = 4\delta(f) - \frac{5}{2}[\delta(f-500) + \delta(f+500)]$



$Y(f) = 200\,[\delta(f-5000) + \delta(f+5000)] - 125[\delta(f-4500) + \delta(f+4500) + \delta(f-5500) + \delta(f+5500)]$



$P_y = \frac{A_c^2}{2} P_m = \frac{100^2}{2}\left(4^2 + 5^2\,\frac{1}{2}\right) = 142\,500\ W = 142{,}5\ kW$

b) $y(t) \rightarrow (\otimes) \rightarrow$ AGF $\rightarrow z(t)$, $\hat{c}(t) \rightarrow (\otimes)$, $e(t)$, AGF: $BG = B = 500$ Hz

$\hat{c}(t) = 2 + 4\cos\left(\frac{10\,000\pi t}{2\pi\,fc}\right) - 2\cos\left(\frac{30\,000\pi t}{2\pi\,3fc}\right)$

$e(t) = y(t).\hat{c}(t) = 100\,m(t)\cos(2\pi f_c t)\,[2 + 4\cos(2\pi f_c t) - 2\cos(2\pi\,3f_c t)]$

$= 200\,m(t)\cos(2\pi f_c t) + 400\,m(t)\cos^2(2\pi f_c t) - 200\,m(t)\cos(2\pi f_c t)\cos(2\pi\,3f_c t)$

$= 200\,m(t)\cos(2\pi f_c t) + 200\,m(t) + 200\,m(t)\cos(2\pi\,2f_c t) - 100\,m(t)\cos(2\pi\,2f_c t)$

$-100\,m(t)\cos(2\pi\,4f_c t)$

$= 200\,m(t) + 200\,m(t)\cos(2\pi f_c t) + 100\,m(t)\cos(2\pi\,2f_c t) - 100\,m(t)\cos(2\pi\,4f_c t)$

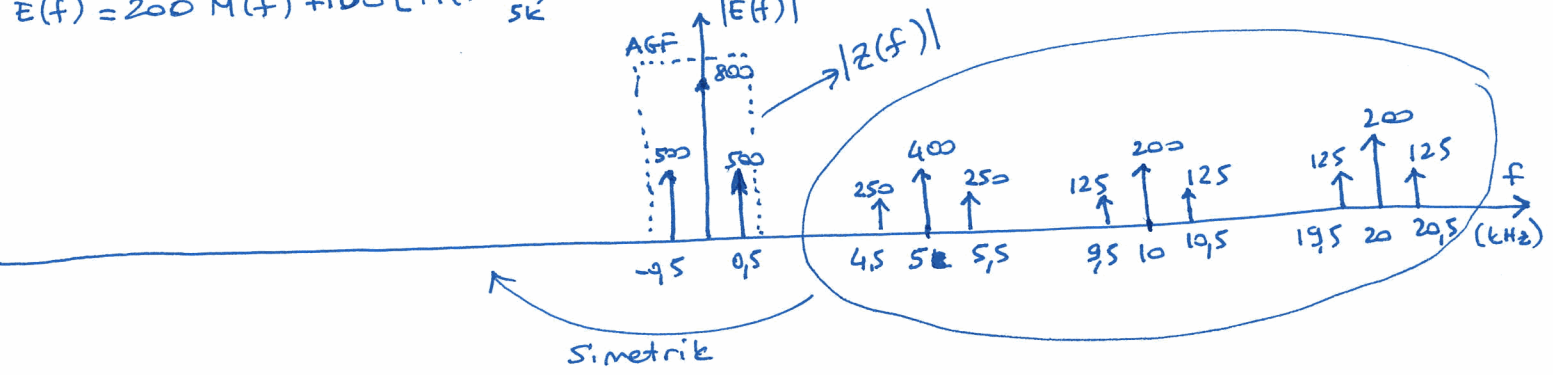
AGF'den geçmet.

$z(t) = 200\,m(t)$ mesaj işareti elde edilmiştir.

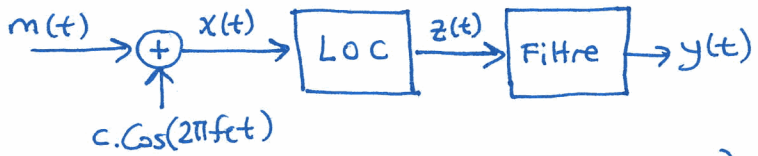
$Z(f) = 200\,M(f)$ ($M(f)$'nin spektrumunda alanlar 200 ile çarpılır.)

$E(f) = 200\,M(f) + 100\,[M(f-f_c) + M(f+f_c)] + 50\,[M(f-10k) + M(f+10k)] - 50[M(f-20k) + M(f+20k)]$

$5k$



(3)



$x(t) = m(t) + c.\cos(2\pi f_c t)$

$z(t) = a\,x(t) + b\,x^2(t)$

$f_c = 100$ kHz

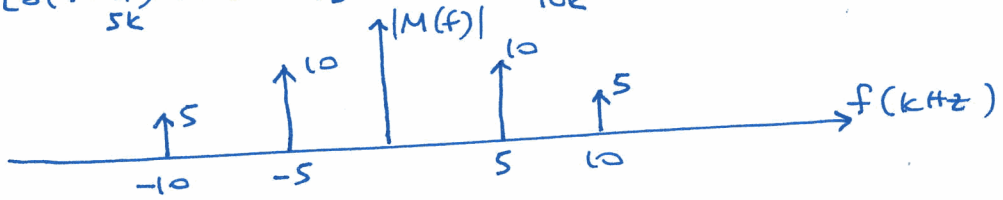
a) $m(t) = 10\cos(2\pi \underset{f_2}{10k}\,t) + 20\cos(2\pi \underset{f_1}{5k}\,t) \rightarrow BG\{m(t)\} = B = 10$ kHz

$a=0,\ b=c=1 \Rightarrow x(t) = m(t) + \cos(2\pi f_c t)$, $z(t) = x^2(t)$

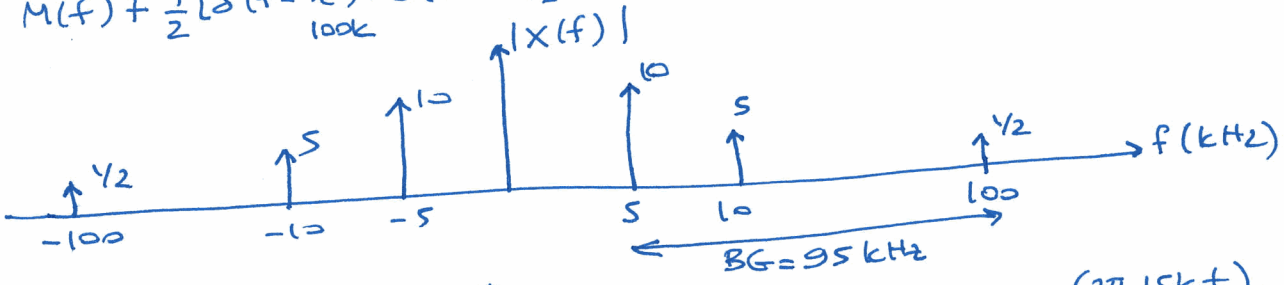
$z(t) = x^2(t) = m^2(t) + \underbrace{\cos^2(2\pi f_c t)}_{\frac{1}{2}[1+\cos(2\pi 2f_c t)]} + \underbrace{2\,m(t)\cos(2\pi f_c t)}_{\text{DSB-SC-AM işareti}}$

$y(t) = 2\,m(t)\cos(2\pi f_c t)$ Filtre: BGF, M.F. $= f_c = 100$ kHz, $BG = 2B = 20$ kHz

$M(f) = 10\,[\delta(f-\underset{5k}{f_1}) + \delta(f+f_1)] + 5\,[\delta(f-\underset{10k}{f_2}) + \delta(f+f_2)]$



$X(f) = M(f) + \frac{1}{2}[\delta(f-\underset{100k}{f_c}) + \delta(f+f_c)]$

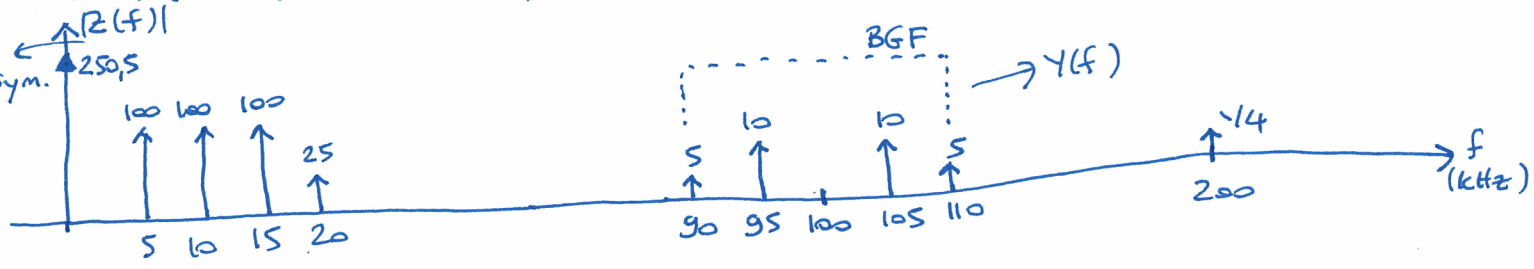


$m^2(t) = 100\cos^2(2\pi \underset{10k}{f_2}t) + 400\cos^2(2\pi \underset{5k}{f_1}t) + \underbrace{200\cos(2\pi\,5k\,t) + 200\cos(2\pi\,15k\,t)}_{u}$

$= 50 + 50\cos(2\pi\,20k\,t) + 200 + 200\cos(2\pi\,10k\,t) +$

$\mathcal{F}\{m^2(t)\} = 250\,\delta(f) + 100\,[\delta(f-5k)+\delta(f+5k)] + 100\,[\delta(f-10k)+\delta(f+10k)] +$
$100\,[\delta(f-15k)+\delta(f+15k)] + 25\,[\delta(f-20k)+\delta(f+20k)]$

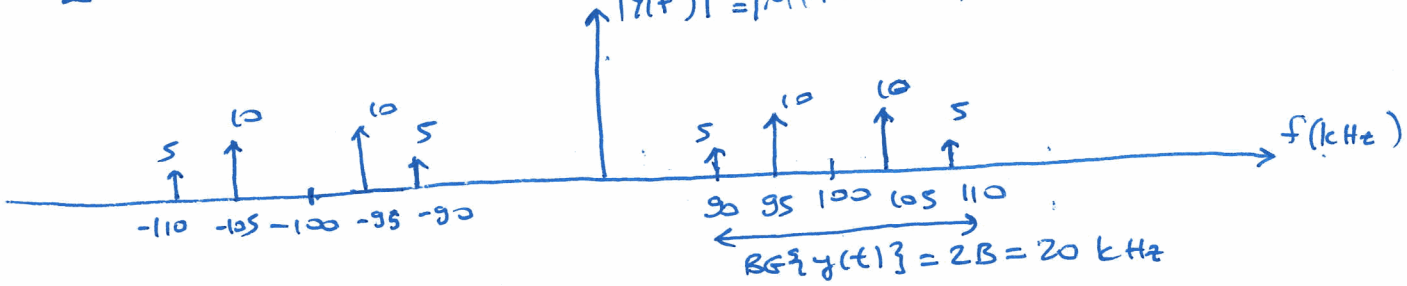
$Z(f) = \mathcal{F}\{m^2(t)\} + \frac{1}{2}\delta(f) + \frac{1}{4}[\delta(f-\underset{2f_c}{200k}) + \delta(f+200k)] + [M(f-\underset{f_c}{100k}) + M(f+100k)]$



$Y(f) = M(f-100k) + M(f+100k)$

$P_y = \frac{A_c^2}{2} P_m = \frac{4}{2}\left[10^2\frac{1}{2} + \frac{20^2}{2}\right] = 500$ W

$|Y(f)| = |M(f-100k) + M(f+100k)|$



b) $a=8$, $b=1$, $c=250$, $m(t)=\cos(2\pi 100t)$

$x(t)=m(t)+250\cos(2\pi 100k t)=\cos(2\pi 100t)+250\cos(2\pi 100k t)$

$z(t)=8x(t)+x^2(t)$

$=8\cos(2\pi 100t)+2000\cos(2\pi 100k t)+\underbrace{\cos^2(2\pi 100t)}_{\frac{1}{2}[1+\cos(2\pi\cdot 200t)]}+2\cos(2\pi 100t)\cdot 250\cos(2\pi 100k t)$

$250^2\underbrace{\cos^2(2\pi 100k t)}_{\frac{1}{2}[1+\cos(2\pi 200k t)]}$

(BGF: MF=100kHz, BG=20kHz)' den temel banddaki ve 200kHz'deki bileşenler geçmez.

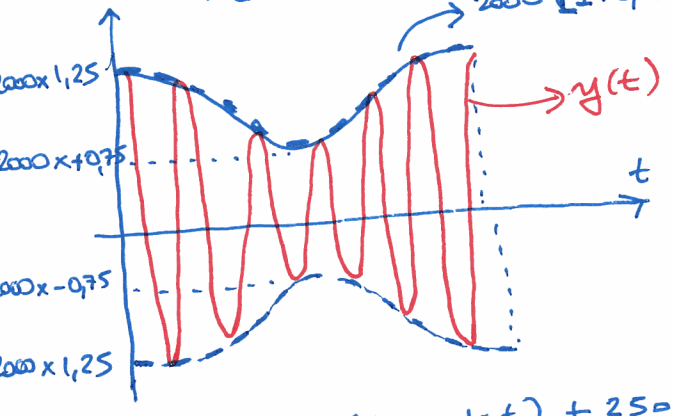
B.T. DSB-SC

$y(t)=2000\cos(2\pi 100k t)+500\cos(2\pi 100t)\cos(2\pi 100k t)$

$=\underbrace{2000}_{A_c}\left[1+\frac{1}{4}\underbrace{\cos(2\pi 100t)}_{m(t)}\right]\cos(2\pi\underbrace{100k}_{f_c}t)$ → Büyük Taşıyıcılı GYB-GM (DSB-LC-AM)

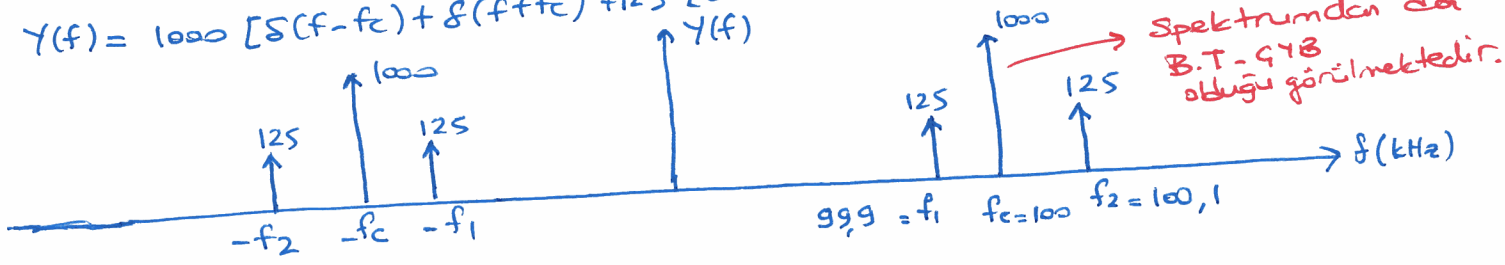
(DSB-SC işaretine büyük taşıyıcı eklenmiş.)

$2000[1+0{,}25\cos(200\pi t)]$



$y(t)=2000\cos(2\pi\underbrace{100k}_{f_c}t)+250\left[\cos(2\pi\underbrace{100{,}1k}_{f_2}t)+\cos(2\pi\underbrace{99{,}9k}_{f_1}t)\right]$

$Y(f)=1000[\delta(f-f_c)+\delta(f+f_c)+125[\delta(f-f_1)+\delta(f+f_1)+\delta(f-f_2)+\delta(f+f_2)]$



Spektrumdan da B.T-GYB olduğu görülmektedir.

$P_y=2000^2\frac{1}{2}+250^2\left(\frac{1}{2}+\frac{1}{2}\right)=2\,062\,500\ W$